

Gas Leakage Detection System Using Arduino

Introduction

Gas leaks can lead to fire, explosions and health hazards. This project demonstrates a simple embedded solution for early gas leak detection.

Problem Statement

Many locations lack affordable gas monitoring systems. A compact Arduino-based detector provides an economical safety solution.

Objectives

Detect combustible gases, provide audio/visual alerts, reduce accident risk, and demonstrate embedded systems.

Existing System

Traditional detectors are expensive and often unavailable for educational use.

Proposed System

An Arduino reads the MQ-2 sensor and controls an LED and buzzer whenever gas exceeds the threshold.

Hardware Components

Arduino UNO, MQ-2 Sensor, LED, Piezo Buzzer, Breadboard, 220Ω Resistor, Jumper Wires, USB Cable.

Software

Arduino IDE using Embedded C/C++.

Circuit Connections

MQ-2 VCC→5V, GND→GND, D0→D2. LED Anode→D8 through 220Ω resistor, Cathode→GND. Buzzer Positive→D9, Negative→GND.

Methodology

Initialize hardware, warm up the MQ-2 sensor, monitor sensor output continuously, activate alarm when gas is detected, return to monitoring when safe.

Working Principle

The MQ-2 detects combustible gases. Arduino reads the digital output and immediately switches ON the LED and buzzer if gas is detected.

Results

The prototype successfully detected alcohol vapour/LPG test conditions and generated immediate alerts.

Applications

Homes, kitchens, hotels, laboratories, industries, LPG storage and educational demonstrations.

Advantages

Low cost, easy to build, portable, reliable and suitable for continuous monitoring.

Limitations

Detects only supported gases, requires sensor warm-up and stable power.

Future Enhancements

ESP32 IoT monitoring, GSM alerts, cloud dashboard, automatic exhaust fan, automatic gas shutoff valve.

Conclusion

The project successfully demonstrates a practical embedded safety application using Arduino and the MQ-2 gas sensor.

References

Arduino Documentation, MQ-2 Datasheet, Arduino IDE Documentation.